What is specific for message passing IPC (inter-process communication) mechanisms?

Select one:

- a. They use syscalls like `read()`, `write()`, `recv()`, `send()`.
- b. They use an userspace buffer.
- c. all apply
- d. They are faster than shared memory IPCs.

Clear my choice

Why is modularity an important concept in designing an application?

Select one:

- a. All apply.
- b. Modular code is easier to debug.
- c. A modular application is easier to maintain.
- d. Modular code is reusable.

Clear my choice

UNIX sockets, pipes and FIFOs are:

Select one:

a. inter-process communication mechanisms that do not use memory.

b. inter-process communication mechanisms based on message transfer that use buffers in kernel space and are accessed through file descriptors.

c. inter-process communication mechanisms based on shared memory in user space.

d. inter-process communication mechanisms based on message transfer that use buffers in user space and are accessed through file descriptors.

Clear my choice

What is an advantage of using processes instead of threads when we want to modularize an application?

Select one:

- a. There is no advantage of using processes over threads.
- b. Processes can run on multiple CPUs, but threads cannot.
- c. Processes can share memory more easily than threads using the `mmap()` syscall with the `MAP_SHARED` flag.
- d. Processes are a better way of confinement due to individual Virtual Address Spaces.

Clear my choice

What is the difference between a pipe and a FIFO?

Select one:

- a. There is no difference between a pipe and a FIFO
- b. Pipes can only be used with related processes, while FIFOs can also be used with unrelated processes.
- c. Pipes are implemented in userspace, and FIFOs are implemented in kernel space.
- d. Pipes use file descriptors, and FIFOs use signals.

Clear my choice

What is an advantage of using processes instead of threads when we want to modularize an application?

Select one:

- a. Processes can share memory more easily than threads using the `mmap()` syscall with the `MAP_SHARED` flag.
- b. Processes are a better way of confinement due to individual Virtual Address Spaces.
- c. There is no advantage of using processes over threads.
- d. Processes can run on multiple CPUs, but threads cannot.

Clear my choice

What is an advantage of using threads instead of processes when we want to modularize an application?

Select one:

- a. Threads are faster than processes.
- b. Interaction between threads is easier because they share the VAS (Virtual Address Space) of a process.
- c. Interaction between threads is easier because they use TLS (Thread Local Storage).
- d. There is no advantage of threads over processes.

Clear my choice

Which is the most appropriate definition for signals?

Select one:

- a. A signal is an asynchronous notification sent to a process.
- b. A signal is an interrupt sent by the MMU (Memory Management Unit).
- c. A signal is a mechanism to kill a process.
- d. A signal is a synchronization mechanism.

Clear my choice

Why is a signal considered an asynchronous operation?

Select one:

- a. Because there is no synchronization mechanism between the signal and the operating system.
- b. Because signals use asynchronous socket operations.
- c. Signals are not asynchronous.
- d. Because we register a handler function and this is called when the signal is received.

Clear my choice

A Similarity between an UNIX socket and a file is:

Select one:

- a. Both UNIX sockets and files are backed-up by storage (e.g disk).
- b. Both UNIX sockets and files are for persistent storage.
- c. Both UNIX sockets and files have an entry in the file system.
- d. Both UNIX sockets and files are slow inter-process communication mechanisms.

Clear my choice

What is a difference between UNIX and network (BSD) sockets?

Select one:

- a. BSD sockets are more efficient than UNIX sockets.
- b. BSD sockets use file descriptors, and UNIX sockets do not.
- c. BSD sockets do not use file descriptors, and UNIX sockets do.
- d. BSD sockets can be used for distributed applications while UNIX sockets are used locally.

Clear my choice

What is a difference between UNIX and network (BSD) sockets?

Select one:

- a. BSD sockets can be used for distributed applications while UNIX sockets are used locally.
- b. BSD sockets do not use file descriptors, and UNIX sockets do.
- c. BSD sockets use file descriptors, and UNIX sockets do not.
- d. BSD sockets are more efficient than UNIX sockets.

Clear my choice

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Can we use BSD sockets for local communication?

Select one:

- a. No. Because BSD sockets are network sockets and can only be used for remote communication.
- b. Yes. Only on port 80.
- c. Yes. Using any network interface.
- d. Yes. Using BSD sockets bound to local files.

Clear my choice

Can a pipe be used after `execve()` has been called in the child process?

Select one:

- a. Yes, because the `execve()` system call does not close file descriptors by default.
- b. No, because the `execve()` system call closes all file descriptors except STDIN, STDOUT and STDERR.
- c. No, because the `execve()` system call changes the entire Virtual Address Space.
- d. No, because the `execve()` system call closes all file descriptors.

Clear my choice

UNIX sockets, pipes and FIFOs are:

Select one:

- a. inter-process communication mechanisms based on message transfer that use buffers in user space and are accessed through file descriptors.
- b. inter-process communication mechanisms based on message transfer that use buffers in kernel space and are accessed through file descriptors.
- c. inter-process communication mechanisms that do not use memory.
- d. inter-process communication mechanisms based on shared memory in user space.

Clear my choice

Can we use semaphores as an inter-thread communication mechanism?

Select one:

- a. Yes. Because we can use the semaphore for exclusive access to a shared resource.
- b. No. Because we cannot use the `wait()` and `post()` methods in two different threads.
- c. Yes. Because we can use the `wait()` and `post()` methods in two different threads.
- d. No. Because we cannot send text messages using semaphores.

Clear my choice

What is an advantage of using threads instead of processes when we want to modularize an application?

Select one:

- a. Threads are faster than processes.
- b. There is no advantage of threads over processes.
- c. Interaction between threads is easier because they use TLS (Thread Local Storage).
- d. Interaction between threads is easier because they share the VAS (Virtual Address Space) of a process.

Clear my choice

What is the most efficient method of inter-process communication?

Select one:

- a. Message Passing.
- b. Signals.
- c. Reading/Writing to files.
- d. Shared Memory.

Clear my choice

Can we use shared memory between two unrelated processes?

Select one:

- a. Yes. By using file mappings.
- b. Yes. By using sockets.
- c. No.
- d. Yes. By using anonymous mappings.

Clear my choice

A difference between an UNIX socket and a file is:

Select one:

- a. UNIX sockets file descriptors are not kept in the process's file descriptor table, while files file descriptors are.
- b. The Unix socket is backed up by memory, and the file is usually backed up by storage.
- c. The UNIX socket is used for network communication, and the file is used to store data.
- d. The UNIX socket can also be used in distributed systems, but files can only be used locally.

Clear my choice

Can we use BSD sockets for local communication?

Select one:

a. Yes. Using any network interface.

b. Yes. Using BSD sockets bound to local files.

c. Yes. Only on port 80.

d. No. Because BSD sockets are network sockets and can only be used for remote communication.

Clear my choice

Can we use UNIX sockets for remote communication?

Select one:

- a. Yes. Using the loopback interface.
- b. Yes. Because UNIX sockets are network sockets.
- c. No. Because UNIX sockets only store data in a local kernel buffer.
- d. Yes. Because UNIX sockets can bind to the network interface.

Clear my choice

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Can we use semaphores as an inter-process communication mechanism?

Select one:

- a. Yes. By using a named semaphore.
- b. No. Because semaphores can be used only for threads.
- c. Yes. By passing the virtual address of the semaphore through a pipe when creating a new process.
- d. Yes. But only for related processes.

Clear my choice

For a local communication what is better to use between UNIX and BSD sockets?

Select one:

- a. BSD sockets because they use the loopback interface and UNIX sockets use files.
- b. BSD sockets because they are faster than UNIX sockets.
- c. UNIX sockets because we can't use BSD sockets locally.
- d. UNIX sockets because they are faster than BSD sockets.

Clear my choice

For a local communication what is better to use between UNIX and BSD sockets?

Select one:

- a. BSD sockets because they use the loopback interface and UNIX sockets use files.
- b. UNIX sockets because we can't use BSD sockets locally.
- c. UNIX sockets because they are faster than BSD sockets.
- d. BSD sockets because they are faster than UNIX sockets.

Clear my choice

Can we use UNIX sockets for remote communication?

Select one:

- a. No. Because UNIX sockets only store data in a local kernel buffer.
- b. Yes. Using the loopback interface.
- c. Yes. Because UNIX sockets can bind to the network interface.
- d. Yes. Because UNIX sockets are network sockets.

Clear my choice